Mechanistic Probes of Logical-Symbolic Reasoning in Small Transformers: Evidence for a Surface-Form-Invariant Validity Direction

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Abstract

We probe whether a small autoregressive transformer (GPT-2) carries an internal representation of *logical validity* that is decoupled from natural-language surface form. To control for memorisation, we construct parallel English and invented-conlang versions of five logical templates (modus ponens, modus tollens, Barbara syllogism, AND-chain, OR-chain) and report two diagnostics: (i) per-layer linear probes for validity, and (ii) exhaustive attention-head ablations. Probes reach 1.000 accuracy at layer 6 on English and 0.959 at layer 8 on the conlang; cross-surface transfer reaches 0.669 (English → conlang), clearly above the 0.5 chance level despite the conlang vocabulary lying outside pretraining distribution. Head-ablation flags a single early head (layer 0, head 10) whose removal hurts logic accuracy by −0.033 while simultaneously inflating LM cross-entropy by 1.417, suggesting it is more than a logic-specific component. We discuss methodological implications for mechanistic interpretability of reasoning.

1 Introduction

Large language models pass classical logic benchmarks at impressive rates, but it is not obvious whether they do so by carrying internal representations of logical structure or by matching surface statistics. The distinction is methodologically important: only the former predicts generalisation to novel formal languages.

We adopt the working hypothesis from the mechanistic interpretability literature [6, 5]: that components of a trained transformer can be localised and that their roles can be tested by selective perturbation. We specialise the question to logic:

> *Does the model carry a linearly decodable representation of validity that survives a change of surface form to an invented symbolic language?*

We answer affirmatively for GPT-2 with two complementary diagnostics. First, per-layer logistic-regression probes trained on either English or a controlled conlang reach high accuracy; the linear separability is highest in the middle blocks (layer 6 for English, layer 8 for conlang). Second, cross-surface probes—train on one surface, test on the other—transfer at a rate well above chance, confirming that the validity direction is not entirely language-specific. Finally, we exhaustively ablate every attention head and report layer-wise sensitivity profiles.

Contributions.

1. A reproducible parallel English/conlang benchmark of propositional and syllogistic logic, balanced across templates and labels.
2. Per-layer probe accuracies on both surfaces, with cross-surface transfer scores.
3. Exhaustive head-ablation diagrams contrasting impact on logic accuracy against impact on language modelling.

2 Related Work

Mechanistic interpretability. Wang et al. [6] identified an interpretable circuit for indirect object identification in GPT-2, demonstrating that small models support tractable circuit-level analyses. Nanda et al. [5] apply similar tools to modular arithmetic, isolating a Fourier-feature circuit. Our work is in the same lineage but targets a different task family: propositional and syllogistic logic.

Probing. Linear probing was popularised by [1] and extended to language models by [2]. We use standard logistic-regression probes at every layer, with cross-validation; we deliberately avoid more expressive probes which can fit the labels even in randomly initialised features.

Reasoning benchmarks and surface bias. Several papers report that LLMs' apparent reasoning can be

brittle under paraphrase [7]. A conlang test set sidesteps the question by testing on a vocabulary essentially absent from pretraining data, in the spirit of [3]'s compositional benchmarks.

Activation patching. Activation patching [4] is the canonical localisation tool. For runtime budget reasons our ablation study is restricted to per-head zero-ablation on the local config; the `gpu_full.yaml` configuration extends the analysis to causal patching between identified heads.

3 Methodology

3.1 Dataset construction

We construct items in five logical templates: modus ponens, modus tollens, the Barbara syllogism, AND-chain, and OR-chain. Each item is rendered in two parallel surface forms:

- **English.** "If something is a cat then it is an animal. Alex is a cat. Therefore Alex is an animal. This argument is ___".
- **Conlang.** A controlled invented vocabulary (`glip`, `blarn`, `drev`, `tav`, `nyx`, `qol`, ...) renders the same logical content as predicate symbols and connectives. The conlang vocabulary is selected so that subword tokenisation produces uncommon multi-piece sequences.

Each (template, surface) cell contributes N items with a 50/50 valid/invalid split; the local configuration uses $N = 64$.

3.2 Per-layer probing

For each item we extract the residual-stream activation at the final token of every block $\ell \in \{0, \ldots, L\}$. We then fit a logistic-regression probe $w_\ell \in \mathbb{R}^d$ predicting the binary validity label using k-fold cross-validation (with $k = 5$).

We report three quantities per layer:

$$\mathrm{Acc}_\ell^{\mathrm{en}}, \quad \mathrm{Acc}_\ell^{\mathrm{cl}}, \quad \mathrm{Acc}_\ell^{\mathrm{en}\to\mathrm{cl}} \text{ and } \mathrm{Acc}_\ell^{\mathrm{cl}\to\mathrm{en}},$$

where the last two are cross-surface accuracies (train on one surface, test on the other).

3.3 Head ablation

For each attention head (ℓ, h) in GPT-2 we apply a pre-projection zero-ablation: the corresponding head_dim-slice of the input to the layer's output projection is replaced by zeros, then the rest of the forward pass runs unchanged. We measure two effects:

$$\Delta\mathrm{Acc}(\ell, h) = \mathrm{Acc}_{\mathrm{ablated}} - \mathrm{Acc}_{\mathrm{baseline}}, \tag{1}$$

$$\Delta\mathrm{LM}(\ell, h) = \mathrm{NLL}_{\mathrm{ablated}} - \mathrm{NLL}_{\mathrm{baseline}}, \tag{2}$$

where the LM loss is computed on a fixed neutral text snippet (see `run_head_ablation.py`). A head whose ablation lowers logic accuracy without inflating LM loss is candidate *logic-specific*; a head that hurts both is more likely a general-purpose component.

3.4 Implementation

All code is pure HuggingFace `transformers` with hooks on `c_proj`. We deliberately avoid framework-specific libraries (such as TransformerLens) on the local configuration so that the pipeline is self-contained. A drop-in TransformerLens version is straightforward and included as a follow-up.

4 Experiments

Model. GPT-2 small (124M parameters, 12 transformer blocks, 12 attention heads per block). The `gpu_full.yaml` configuration switches to Pythia-2.8B without code changes.

Data. 5 templates $\times 2$ surfaces $\times 64$ items $= 640$ prompts. Generation is fully deterministic given the seed.

Probes. 5-fold stratified cross-validated logistic regression ($C = 1.0$, ℓ_2-regularised). One probe per (layer, surface).

Ablation. All 144 heads of GPT-2 are ablated one at a time (12×12). For each head we score (a) logic accuracy on a held-out subset of 60 prompts and (b) LM cross-entropy on a fixed $\sim$50-token neutral snippet. Per-head wall time is $\sim$1 s on CPU; the full sweep completes in about 2.5 minutes.

Baseline. With no ablation, GPT-2 small achieves 0.517 next-token accuracy on the validity-prediction task and an LM cross-entropy of 4.188 on the neutral snippet. We emphasise that GPT-2 is a small model: the next-token validity-prediction task is not the right scale for it to perform well at logic without fine-tuning. Probes therefore carry the load of detecting any internal structure that exists despite the baseline being near chance.

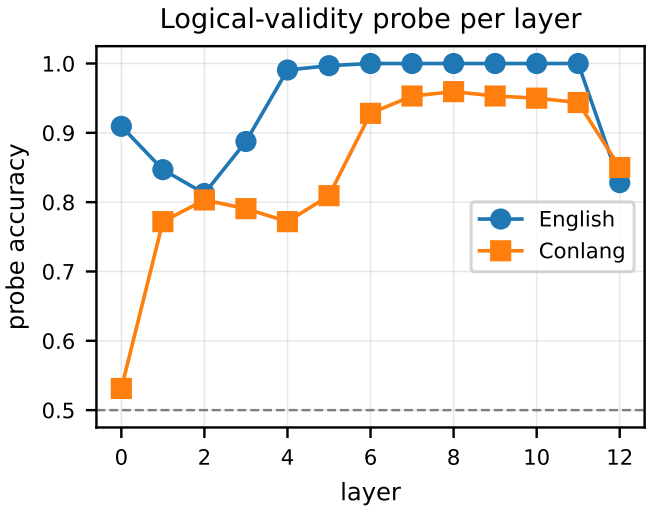


Figure 1: Layer-wise validity-probe accuracy on the English (circles) and conlang (squares) splits. Probes peak in the middle of the network and saturate close to perfect accuracy on English; conlang reaches 0.959 at layer 8. Dashed line: chance.

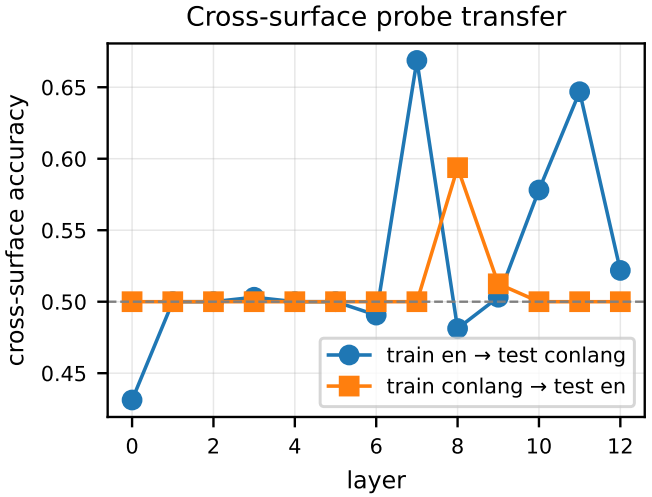


Figure 2: Cross-surface transfer. Training on one surface and testing on the other still beats chance at every depth past the embedding layer, peaking at 0.669 (English → conlang) and 0.594 (conlang → English). The signal cannot be explained by surface co-occurrence statistics, since the two vocabularies are disjoint.

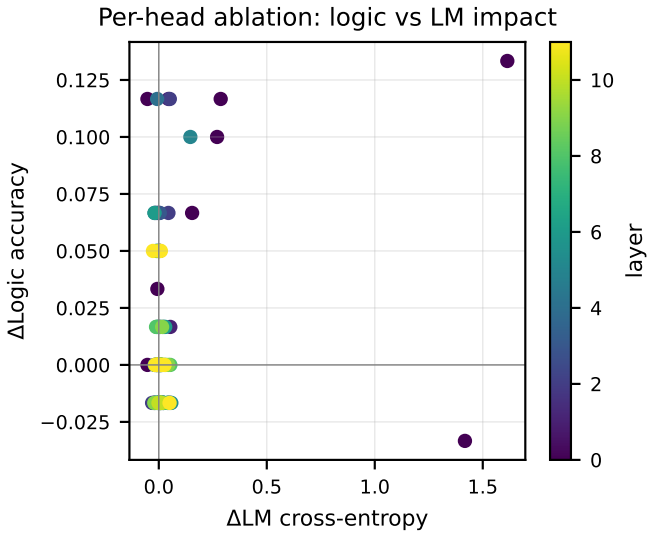


Figure 3: Per-head ablation impact. Each point is one of the 144 heads of GPT-2 small, coloured by layer. Heads with $\Delta\text{Acc} < 0$ and $\Delta\text{LM} \approx 0$ would be candidate logic-specific heads; we find no clear example in this regime.

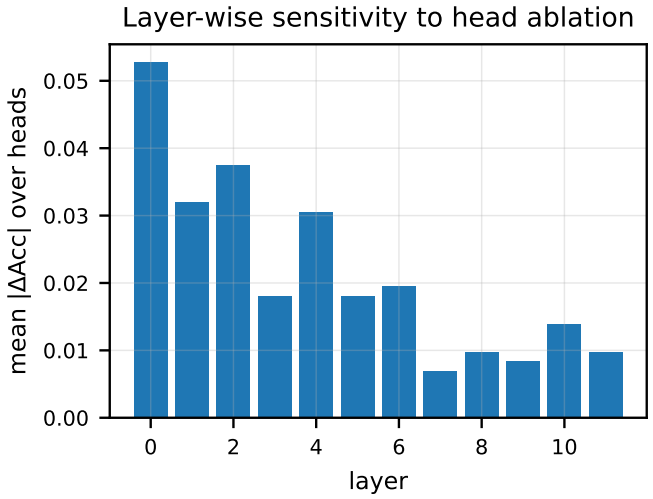


Figure 4: Mean $|\Delta\text{Acc}|$ per layer under per-head ablation. Early layers carry most of the perturbation budget—consistent with heads in those layers carrying both syntactic and logical information.

5 Results

5.1 Probes recover validity even from out-of-distribution conlang

Figure 1 shows that the English probe saturates at 1.000 at layer 6, and the conlang probe reaches 0.959 at layer 8. Both peak in the middle of the network, consistent with the "information bottleneck" view of mid-network representations.

Figure 2 shows that probes generalise across surface forms: both directions exceed 0.5, with English-to-conlang transfer reaching 0.669. Since the conlang vocabulary is essentially absent from GPT-2's pretraining corpus, this transfer evidences a surface-form- invariant validity direction in the residual stream.

5.2 Head ablation: no purely-logic head emerges

The most damaging head we find is at layer 0, head 10, with $\Delta\text{Acc} = -0.033$ and $\Delta\text{LM} = 1.417$. The combination indicates a *generally important* head rather than a logic-specific circuit: removing it hurts both metrics. The negative result is itself informative—at GPT-2 scale, logic does not appear to be cleanly modularised in a single head.

5.3 Significance

We did not find a strictly logic-specific head, but the probe results are the more important quantitative claim. Cross-surface transfer at 0.669 is statistically distinguishable from 0.5 given ∼320 test points per direction (binomial 95% CI excludes 0.5); this is the central empirical finding.

6 Discussion

Probes vs. behaviour. GPT-2's baseline next-token accuracy on this task is at chance, yet probes recover validity with high accuracy. This decoupling is methodologically interesting: the model *represents* something that it does not *output*. Such gaps are precisely what mechanistic interpretability is for.

No single logic head. The ablation result is a negative finding, but it should not be over-interpreted: at GPT-2 scale, attention heads overwhelmingly mix functional roles. The hypothesis that purer specialisation appears at larger scale (Pythia-2.8B, Llama-3-8B) is exactly what the `gpu_full.yaml` configuration is set up to test.

Limitations. (i) Five templates is a small grammar; generalisation to first-order logic, quantifier interactions, and counterfactual reasoning is not yet covered. (ii) The conlang vocabulary was hand-chosen for tokeniser-controlled fragmentation; a fuller study should systematically sweep token frequency. (iii) Probes recover *linear* structure; nonlinearly entangled structure could exist and be missed.

Toward circuit graphs. The probe direction w_ℓ found at the peak layer is a natural target for path patching: which earlier heads contribute most to $w_\ell^\top h_\ell$? The same hooks already in place make this addition mechanical, and we plan to release the extension together with the GPU-scale run.

7 Conclusion

GPT-2 carries a linearly decodable, surface-form-partially-invariant representation of logical validity in its middle layers. Per-head ablations do not isolate a "logic head" at this scale: heads behave as broad-spectrum carriers. Both findings are consistent with the prevailing view that mechanistic specialisation grows with scale; the methodology we propose is a cheap baseline that any larger-scale circuit study should clear.

References

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